THE HIGHER DERIVED LOWER SHRIEK ALONG A FIBRED CATEGORY AGREES WITH THE EXPLICITLY CONSTRUCTED FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the delta-functor setup, the computation of the presheaves F_{U'} and of their restrictions to the fibre categories, the resulting vanishing of the higher functors, the quotient argument giving universality, and the final identification with the derived lower shriek.

Editorial note. The label lemma-compute-left-derived-pi-shriek names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source sites-cohomology.tex –label lemma-compute-left-derived-pi-shriek –start 10614 –end 10675 –local-id 3548 returned RESULT=OK (L3548 free, occupancy FREE). Candidate taken from the Lead-forwarded sorted list lane-s4/halfB-priority.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The explicitly constructed functors L_n on abelian presheaves. Same chapter, the lemma constructing the functors L_n by the cosimplicial procedure, whose agreement with the derived lower shriek is the assertion. Supplies the functors whose agreement with L_n pi_! is proved.
- Vanishing of the higher homology of the localisation-by-zero presheaves. Same chapter, the example computing the higher homology of the presheaves j_{!}Z over a localized site, and Modules on Sites, the lemma allowing a presheaf to be written as a quotient of one with vanishing higher L_n. Supplies the effacing objects needed for universality.
- Universality and uniqueness of universal delta functors. Homology, the lemma that effaceability implies universality and the lemma on uniqueness of universal delta functors; Derived Categories, the lemma comparing a right derived functor with a delta functor; Same chapter, the agreement in degree zero. Supplies the final identification of the two families of functors.

Source reference. Original definition 02XN: sources/stacks-chow/tags.txt, source lines 3846–3846. Prerequisite pointer invoked at proof line(s) 10646; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 05TE: sources/stacks-chow/tags.txt, source lines 7478–7478. Prerequisite pointer invoked at proof line(s) 10674; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08PG: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 10044–10044. Prerequisite pointer invoked at proof line(s) 10657; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 010T: sources/stacks-chow/tags.txt, source lines 1339–1339. Prerequisite pointer invoked at proof line(s) 10667; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 010U: sources/stacks-chow/tags.txt, source lines 1340–1340. Prerequisite pointer invoked at proof line(s) 10672; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08PI: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 10526–10526. Prerequisite pointer invoked at proof line(s) 10620; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08PE: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9960–9960. Prerequisite pointer invoked at proof line(s) 10670; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03EW: sources/stacks-chow/tags.txt, source lines 4471–4471. Prerequisite pointer invoked at proof line(s) 10662; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 03EJ: sources/stacks-chow/tags.txt, source lines 4458–4458. Prerequisite pointer invoked at proof line(s) 10636; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08P8: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9701–9701. Prerequisite pointer invoked at proof line(s) 10616; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Assumptions and notation as in Situation 08P8. For $\mathcal{F}$ in $Ab(\mathcal{C})$ and $n \geq 0$ the sheaf $L_n\pi_!(\mathcal{F})$ is equal to the sheaf $L_n(\mathcal{F})$ constructed in Lemma 08PI.*

Proof. Consider the sequence of functors $\mathcal{F} \mapsto L_n(\mathcal{F})$ from $PAb(\mathcal{C}) \to Ab(\mathcal{C})$. Since for each $V \in \mathrm{Ob}(\mathcal{D})$ the sequence of functors $H_n(\mathcal{C}_V, -)$ forms a δ-functor so do the functors $\mathcal{F} \mapsto L_n(\mathcal{F})$. Our goal is to show these form a universal δ-functor. In order to do this we construct some abelian presheaves on which these functors vanish.

For $U' \in \mathrm{Ob}(\mathcal{C})$ consider the abelian presheaf $\mathcal{F}_{U'} = j^{PAb}_{U'!}\mathbf{Z}_{U'}$ (Modules on Sites, Remark 03EJ). Recall that

$$\mathcal{F}_{U'}(U) = \bigoplus\nolimits_{\mathrm{Mor}_{\mathcal{C}}(U,U')} \mathbf{Z}$$

If U lies over $V = p(U)$ in $\mathcal{D}$) and U' lies over $V' = p(U')$ then any morphism $a : U \to U'$ factors uniquely as $U \to h^*U' \to U'$ where $h = p(a) : V \to V'$ (see Categories, Definition 02XN). Hence we see that

$$\mathcal{F}_{U'}|_{\mathcal{C}_V} = \bigoplus\nolimits_{h \in \mathrm{Mor}_{\mathcal{D}}(V,V')} j_{h^*U'!}\mathbf{Z}_{h^*U'}$$

where $j_{h^*U'} : Sh(\mathcal{C}_V/h^*U') \to Sh(\mathcal{C}_V)$ is the localization morphism. The sheaves $j_{h^*U'!}\mathbf{Z}_{h^*U'}$ have vanishing higher homology groups (see Example 08PG). We conclude that $L_n(\mathcal{F}_{U'}) = 0$ for all $n > 0$ and all U'. It follows that any abelian presheaf $\mathcal{F}$ is a quotient of an abelian presheaf $\mathcal{G}$ with $L_n(\mathcal{G}) = 0$ for all $n > 0$ (Modules on Sites, Lemma 03EW). Since $L_n(\mathcal{F}) = L_n(\mathcal{F}^\#)$ we see that the same thing is true for abelian sheaves. Thus the sequence of functors $L_n(-)$ is a universal delta functor on $Ab(\mathcal{C})$ (Homology, Lemma 010T). Since we have agreement with $H^{-n}(L\pi_!(-))$ for $n = 0$ by Lemma 08PE we conclude by uniqueness of universal δ-functors (Homology, Lemma 010U) and Derived Categories, Lemma 05TE. $\square$